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Paleomagnetism of the Neoproterozoic diamictites of the Qiaoenbrak formation in the Aksu area, NW China: Constraints on the paleogeographic position of the Tarim Block

Bin Wen, Yong-Xiang Li*, Wenbin Zhu

State Key Laboratory for Mineral Deposits Research (Nanjing University), School of Earth Sciences and Engineering, Nanjing University, 22 Hankou Road, Nanjing 210093, China

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ABSTRACT

The middle to late Neoproterozoic records the transition from the dispersal of supercontinent Rodinia to the subsequent assembly of supercontinent Gondwana. Determining paleogeographic positions of the major cratons during this transition is crucial to understand the displacement histories of these blocks. However, high-quality paleomagnetic results from well-dated rocks during this period remain sparse for the cratons. In this paper, we have carried out a paleomagnetic study of the succession of the Cryogenian Qiaoenbrak formation (Fm) in the Aksu area, NW China, which contains diamictites that are of probable glacial origin. The age of the studied section is constrained at around 730 Ma based on the stratigraphy and recent zircon U–Pb chronologic data. Stepwise alternating field (AF) and thermal demagnetizations generally reveal two-component magnetizations. The low temperature (coercivity) component was removed by ~300 °C (~30 mT) and the high temperature (coercivity) component decayed toward the origin, defining the characteristic remanent magnetization (ChRM). Rock magnetic data indicate that magnetite is the major remanence carrier. A negative fold test for an isoclinal soft-sediment fold, the presence of both normal and reversed polarities, and positive conglomerate tests in conjunction with the presence of pristine sedimentary features by microscopic inspections of thin sections suggest that the remanence was acquired shortly after the deposition, i.e. post-depositional remanent magnetization (pDRM), which is very close to the magnetic field at the time of deposition. Fourteen out of 17 sites yield stable ChRMs with a mean direction of $D_s = 246.1°$, $I_s = 30.4°$, $k_s = 14.1$, $\alpha_{95} = 11.0°$ in stratigraphic coordinates, corresponding to a paleopole ($Q = 6$) at $\lambda = 6.3°$S, $\varphi = 17.5°$E, $A_{95} = 9.1°$. Together with geologic constraints, the new paleomagnetic results place the Tarim Block at $16.3 \pm 5.6°$N adjacent to either the west or east of Australia (WOA or EOA) at around 730 Ma, but the WOA position is preferred based on the overall relatively favorable evidence. In addition, since the studied diamictites of the Qiaoenbrak Fm that are of probable glaciogenic origin yield a low paleolatitude, this study may lend support to the "Snowball Earth" hypothesis during the Neoproterozoic times.

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1. Introduction

The Neoproterozoic era saw the dispersal of the supercontinent Rodinia, the subsequent re-amalgamation giving birth to another large landmass Gondwana, and the accompanying large-scale glaciations on the fragmentary landmasses (Hoffman et al., 1998; Schrag and Hoffman, 2001; Hoffman and Schrag, 2002; Torsvik, 2003; Condon et al., 2005; Deynoux et al., 2006; Trindade and Macouin, 2007; Li et al., 2008; Meert and Lieberman, 2008; Rino et al., 2008; Macdonald et al., 2010; Bradley, 2011). Precise reconstruction of the major cratons during this interval is crucial to establish the displacement histories of these cratons and to further understand the associated geodynamic processes within a supercontinent cycle. As one of the important cratons in the supercontinent Rodinia and one of three major blocks in China, the Tarim Block contains a complete Mesoproterozoic to Neoproterozoic succession (Gao et al., 1985; XBGMR, 1993; Xiao et al., 2004; Xu et al., 2005, 2009; Turner, 2010; Zhu et al., 2011), which documented the assembly and breakup of the supercontinent Rodinia, and the major glaciations in the Neoproterozoic (e.g. Xiao et al., 2004; Xu et al., 2005, 2009; Lu et al., 2008). However, the paleogeographic positions of the Tarim Block during this interval remain poorly constrained.

The Tarim Block has been qualitatively positioned adjacent to the northwestern Australia at a high latitude based on the similar tectonostratigraphies of both blocks (Hoffman, 1991; Li et al.,

* Corresponding author. Tel.: +86 25 83597573; fax: +86 25 8368 6106.
E-mail address: yxli@nju.edu.cn (Y.-X. Li).

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1996, 2008; Li and Powell, 2001) or the southeastern Australia at a low latitude based on proposed radiating dyke swarms (Lu et al., 2008). Paleomagnetic studies of the Neoproterozoic rocks provide crucial quantitative constraints on the paleopositions of the Tarim Block (Chen et al., 2004; Huang et al., 2005; Zhan et al., 2007). Chen et al. (2004) obtained a paleopole from the Aksu dyke swarms and placed the Tarim Block to the north of Australia at an intermediate latitude of 43 ± 6°N at ~800 Ma. Huang et al. (2005) conducted a paleomagnetic study of the Baiyisi volcanic rocks in the Quruqtagh area and obtained an equatorial paleolatitude for the Tarim Block at ~740 Ma. However, it is not possible that both paleopoles from these two studies can fit the apparent polar wander path (APWP) of Rodinia in one single reconstruction (Evans, 2009). One has to fit one paleopole with the Rodinian APWP and invoke a tectonic or overprinting event to explain the departure of the other pole from the Rodinian APWP (Evans, 2009). The difficulty in reconciling these two poles partly arises from the large age uncertainties of studied rocks. For instance, the Aksu dyke swarms were previously assigned an age of 807 ± 12 Ma (Chen et al., 2004), and was later revised to 785 ± 31 Ma (Zhan et al., 2007). More recently, zircon U–Pb chronologic studies show that the Aksu dyke swarms likely emplaced at ~760 Ma (Zhang et al., 2009a) or even younger than ~730 Ma (Zhu et al., 2011).

The difficulties in reconciling the available paleomagnetic poles emphasize an urgent need to obtain more reliable paleomagnetic data from well-dated strata in the Tarim Block. In this study, we have carried out a detailed paleomagnetic investigation of a succession of the Qiaoenbrak Fm in the Aksu area (Fig. 1a) that contains the Neoproterozoic diamictites, which are of probable glaciogenic origin (Gao et al., 1981, 1985; XBGMR, 1993; Carroll et al., 2001; Gao and Chen, 2003; Zhu and Wang, 2011). The age of the studied section can be fairly well constrained with the new U–Pb zircon ages from the Aksu area and its equivalent stratigraphic interval in the Quruqtagh area of the northeastern Tarim Block (Fig. 1a). This study contributes to constrain the paleogeographic position of the Tarim Block in the Neoproterozoic and could further test the snowball earth hypothesis.

2. Geological setting and sampling

The pre-Cambrian rocks of the Tarim Block mainly crop out along its northern margin and consist of the Neoarchean to Mesoproterozoic metamorphic basement and the Neoproterozoic-Cambrian unmetamorphosed sedimentary cover (Lu et al., 2008; Long et al., 2010; Cao et al., 2011; Shu et al., 2011; Ge et al., 2012; He et al., 2012; Long et al., 2012; Ma et al., 2012; Zhang et al., 2012; Zhao and Guo, 2012; Zhao and Cawood, 2012). In the Aksu area (Fig. 1a), the well-exposed pre-Cambrian successions are composed of, from the bottom to the top, the Proterozoic Aksu Group, the lower Sinian Qiaoenbrak Fm and the Yuermeinak Fm, and the upper Sinian Sugetbrak Fm and the Chigebrak Fm (Fig. 1b). The lower and upper Sinian successions in the Tarim Block correspond to the Cryogenian (Nanhua) and Ediacaran strata, respectively (Fig. 1b) (the Stratigraphy Committee of China, 2001; Gradstein et al., 2004). The Aksu Group consists of metasedimentary rocks, greenschists, and blueschists, and was intruded by a series of NW-trending mafic dykes. The Qiaoenbrak Fm and the Yuermeinak Fm are mainly composed of feldspathic arenite, siltstone, lithic feldspar sandstone, and conglomerates. These two formations also contain diamictites, within which flat-iron stones, striated clasts, and drop stones occur, suggesting that the diamictites were probably of glaciogenic origin (Gao et al., 1981, 1985; XBGMR, 1993). The Sugetbrak Fm is 400–450 m thick and comprises the red conglomerates, fluvial sandstones interbedded with three layers of 15–20 m thick basalt, and the gray lacustrine mudstones. The overlying Chigebrak Fm is composed mainly of thickly bedded, pale gray stromatolitic dolomite which was deposited in an extensive lake or a marine transgressional environment (Gao et al., 1985; Turner, 2010).

We conducted a detailed paleomagnetic investigation of a section in the Qiaoenbrak Fm that contains diamictites in the Aksu area (Fig. 1a and c). The diamictites are represented by two conglomerate beds that are separated by a ~30 m thick sedimentary sequence, which is composed of thinly bedded, greenish siltstones and mudstones and contains a soft-sediment fold (Fig. 1c). Both the lower and the upper conglomerate beds consist of subangular to subrounded, poorly sorted clasts of various lithologic compositions. The lower conglomerate bed is about 20 m thick and is dominated by gray sandstone and red granitic pebbles/cobbles. The upper conglomerate bed is ~50 m thick and contains three 1–2 m thick fine- to coarse-grained sandstone layers that are intercalated within the upper segment of this conglomerate bed. The number of pebbles/cobbles in the uppermost of the conglomerate bed decreases gradually upsection. Below and above the diamictite unit are thin- to medium-bedded red or yellowish sandstones (Fig. 1c).

We collected eight sites from the ~80 m thick red sandstone sequence below the diamictite unit and six sites from the ~90 m thick red sandstone sequence above the diamictite unit. Two sites from the interbedded sandstones within the upper conglomerate bed and one site from the greenish siltstone sequence that separates the two conglomerate beds were collected. Since these beds show little variations in attitudes and generally dip toward 240–250° at about 50–60°, we collected samples from the two conglomerate beds and the soft-sediment fold to perform conglomerate tests and a fold test to constrain the age of remanence (Fig. 1c). Two sites were collected from the two conglomerate beds (sites AK9 and AK18) and about 10 pebbles/cobbles of different lithologies were sampled at each site. For the soft-sediment fold, samples were collected from both limbs and the hinge of the fold (site AK16). All samples were collected with a portable gasoline-powered rock drill and generally 8–12 cores were drilled at each site. The orientation of each core was measured with a magnetic compass mounted on a Pomery orientation device.

3. Laboratory techniques

In the laboratory, cores were cut into standard cylindrical specimens of 2.5 cm in diameter and 2.2 cm in length. To characterize magnetic mineralogy, representative specimens were chosen for several rock magnetic experiments. The isothermal remanent magnetization (IRM) acquisition was measured with an ASC IM-10-30 impulse magnetizer and a JR-6A spinner magnetometer. Spectral analyses of IRM acquisition curves are performed to identify different coercivity components (Kruiver et al., 2001). Thermo-magnetic properties were determined using a KLY-3S kappabridge susceptibility meter coupled with a CS-3 furnace. To further constrain the magnetic mineralogy, representative samples were magnetized sequentially along *Z*-, *Y*-, and *X*-axis at fields of 2.4 T, 0.4 T, and 0.12 T, respectively, and then were subjected to stepwise thermal demagnetization (Lowrie, 1990). The anisotropy of magnetic susceptibility (AMS) was measured using the KLY-3S kappabridge.

Both thermal and alternating filed (AF) demagnetizations were used to isolate the characteristic remanent magnetization (ChRM). Progressive thermal demagnetization was carried out in 16–22 steps at a 50 °C increment for low temperatures (<300 °C) and a 20 or 30 °C increment for high temperatures (>300 °C) up to 600–680 °C. AF demagnetization was conducted in about 18 steps up to 90–100 mT. Since pilot experiments show that thermal demagnetization is more effective in isolating ChRMs than AF demagnetization, the majority of specimens were then subjected to thermal demagnetization. Remanence of specimens

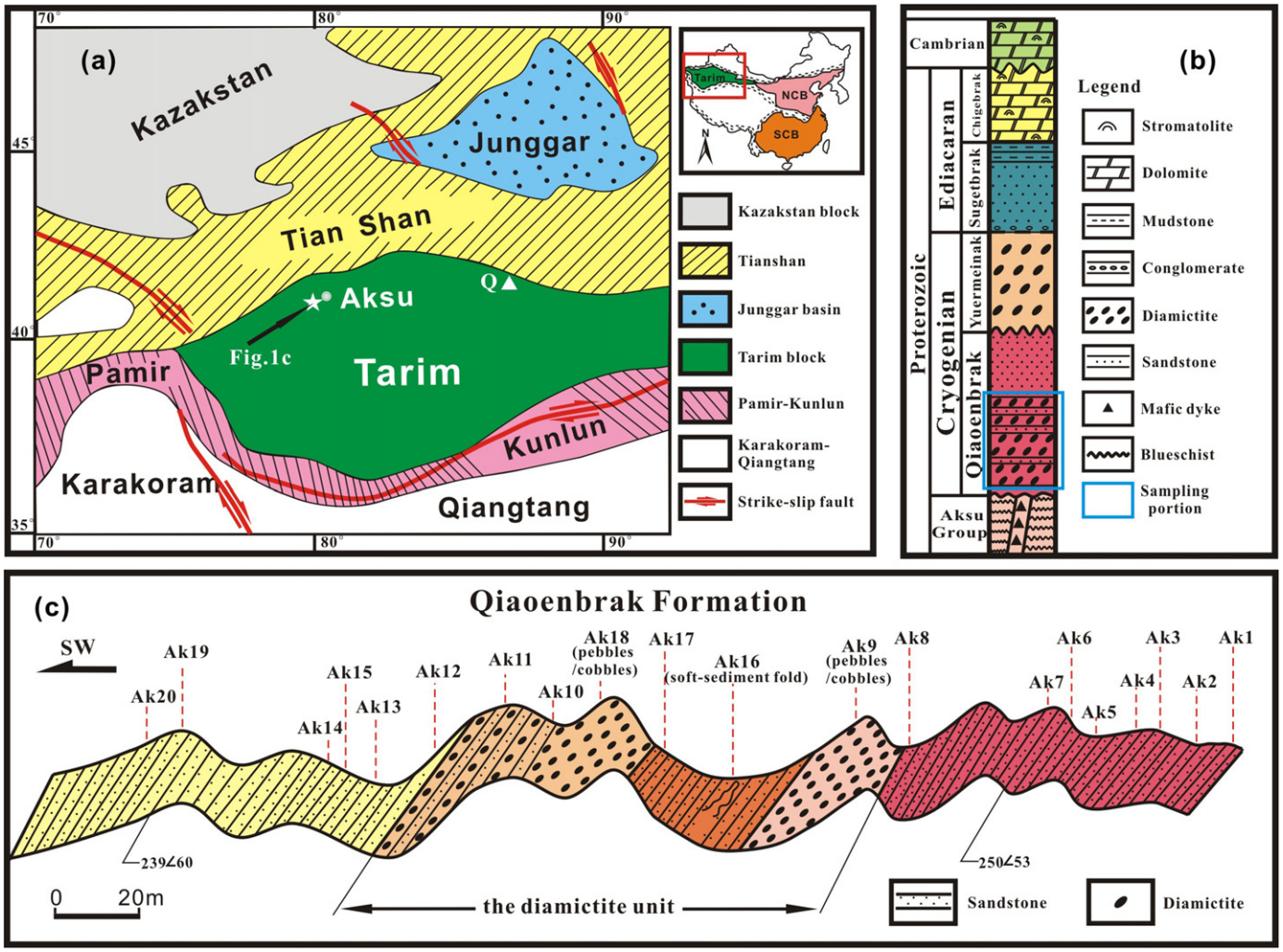


Fig. 1. (a) Sketch map of the Tarim Block and its adjacent areas (after Zhao et al., 2002; Turner, 2010). The study area is located at the northwestern margin of the Tarim Block near Aksu city. Q, Quruqtagh; SCB, South China Block; NCB, North China Block. (b) The Neoproterozoic-Cambrian stratigraphic column of the Aksu area (after Gao et al., 1985; XBGMR, 1993; Turner, 2010). (c) The cross-section of the studied Cryogenian Qiaoenbrak Fm that contains the diamictites. The diamictites are represented by two conglomerate beds. Paleomagnetic samples were collected below, within, and above the diamictite unit at a total of 20 sites (AK1–AK20). 239∠60, the dip direction and dip angle of the strata.

was measured with a 2G Enterprises Inc. cryogenic rock magnetometer (2G-755) housed in a magnetically shielded room in the Paleomagnetism Laboratory of Nanjing University. The magnetic remanence directions were analyzed using the principal component analysis method (Kirschvink, 1980) and the mean directions were calculated with Fisher spherical statistics (Fisher, 1953). The paleomagnetic data were analyzed using the software PMGSC (Version 4.2) of R. Enkin, and some figures were produced using the computer program PaleoMac of Cogné (2003).

In addition, polished thin sections were examined with a microscope to check whether the studied section experienced mineralogic alterations after deposition.

4. Results

4.1. Rock magnetic results

The IRM acquisition curves typically show that a rapid increase in IRM occurs between 0 mT and 200–300 mT of the applied field and full saturation was not reached until up to about 1.2 T (Fig. 2a and b), suggesting that the samples contain predominantly low-coercivity magnetic minerals and minor intermediate to high-coercivity magnetic phases. Spectral analyses of seven representative IRM acquisition curves (Kruiver et al., 2001) reveal that IRMs of the samples can be explained by two coercivity components with the low-coercivity component of 60–100 mT and the intermediate component of 350–630 mT. For samples that have an intermediate component of <400 mT (>400 mT), their low coercivity component typically contributes to more than (less than) 50% of their IRM. Stepwise thermal demagnetization shows that low- and intermediate-coercivity components are dominant and decrease sharply at ~580 °C (Fig. 2c and d), indicating the presence of magnetite. Minor high coercivity component persists up to 690 °C and decays rapidly afterward in some samples (Fig. 2c), suggesting the presence of hematite. Thermal magnetic properties of the samples are characterized by the irreversible feature of magnetic susceptibilities during heating and cooling (Fig. 3), suggesting mineralogic changes during the treatment. Samples from the sandstones below and above the diamictite unit display similar variation patterns of magnetic susceptibility (Fig. 3a, b and d). The subtle decrease in susceptibility from room temperature to ~500 °C may indicate the presence of paramagnetic minerals. Slight increase from ~500 to 570 °C in magnetic susceptibility may result from mineral alteration (probably clay minerals) that

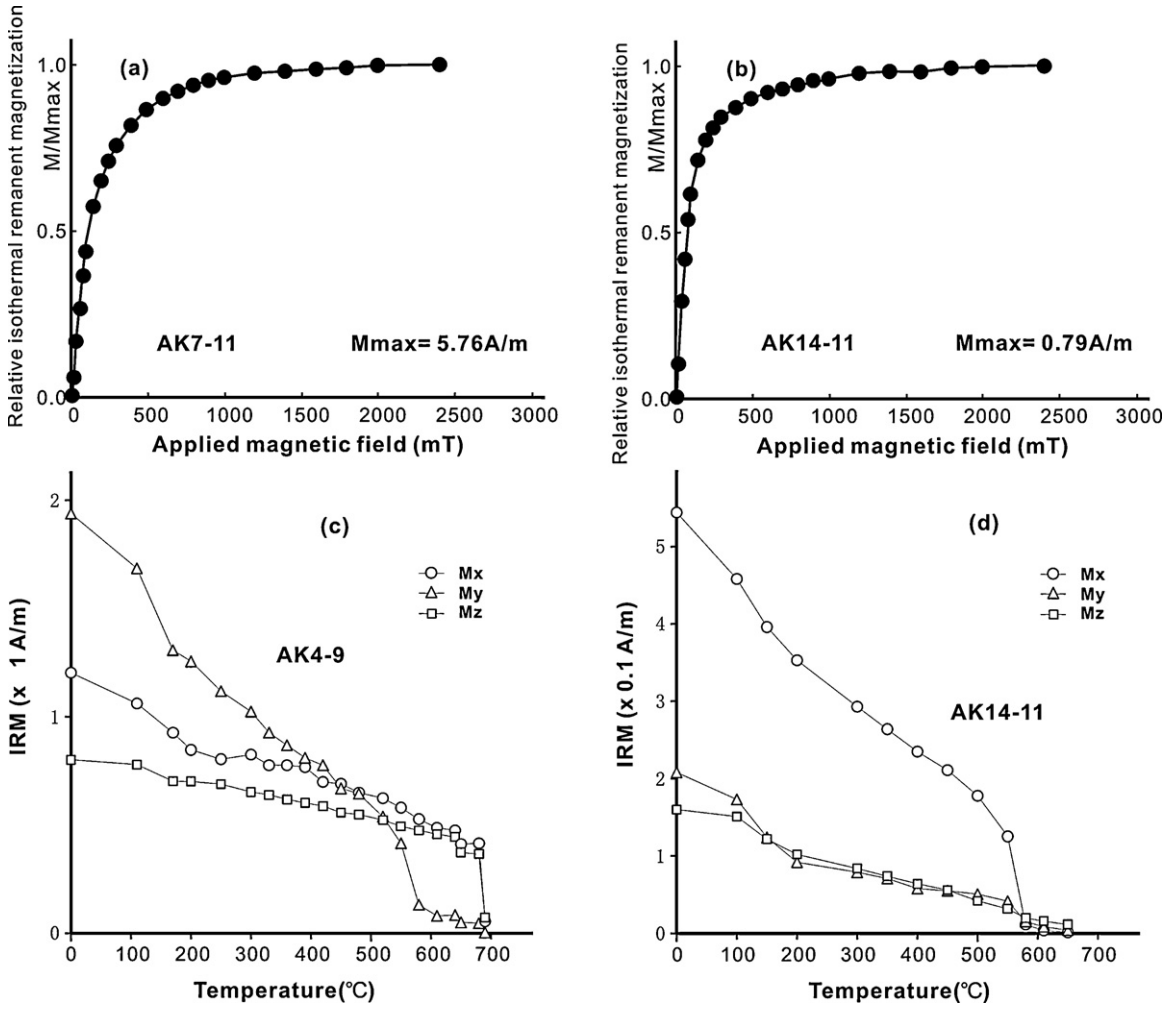


Fig. 2. Isothermal remanent magnetization (IRM) acquisition of representative samples shows that the samples are largely saturated by ~200–300 mT, suggesting that the soft coercivity phases dominate the rocks.

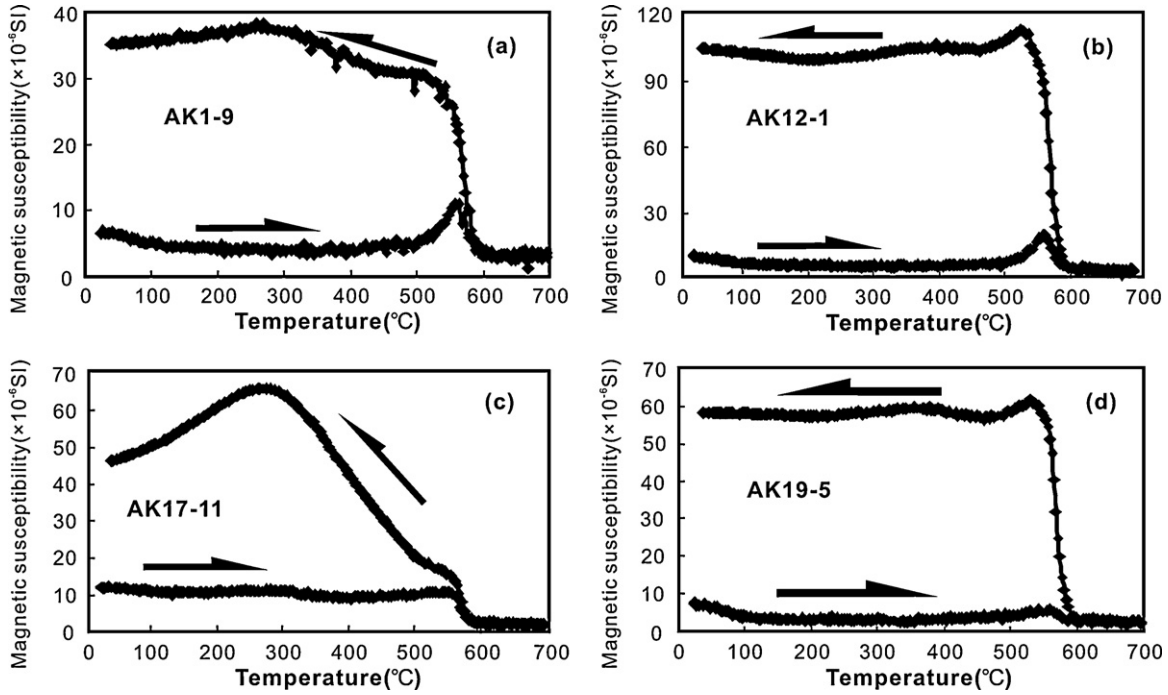


Fig. 3. Thermo-magnetic properties of representative samples in a heating–cooling cycle in the presence of argon gas. The sharp drop at about 580 °C indicates that magnetite is the major remanence carrier in the samples.

produced new magnetite particles. The subsequent relatively rapid decrease from 570 to 600 °C indicates the presence of magnetite. Rapid and large increase in magnetic susceptibility during cooling from ~600 to 570 °C suggests that abundant new magnetite particles were produced (Fig. 3a, b and d). Magnetic susceptibility shows little changes during the subsequent cooling. The sample from siltstones interbedded in the diamictite unit exhibits different thermal magnetic properties (Fig. 3c). Magnetic susceptibility almost remains constant from room temperature to ~580 °C and drops rapidly between 580 °C and 600 °C, indicative of the dominant presence of magnetite. The rapid increase during cooling from 600 to 580 °C, which indicates the production of new magnetite particles, is relatively minor. This is followed by a large and gradual increase and a moderate decrease with peaks at around 250–300 °C, suggesting possible phase changes associated with iron sulfides (Roberts et al., 2011). The above features suggest that magnetite is the major remanence carrier in the samples. AMS data appear to show a weak prolate fabric with the maximum axes perpendicular to the horizontal and the intermediate and minimum axes within the horizontal in geographic coordinates (Fig. 4a). In stratigraphic coordinates, the AMS data show that the maximum axes inclined toward the southwest at ~30° and the intermediate and minimum axes occur in a girdle (Fig. 4b), which may be associated with processes of basal glacial deposits (Gentoso et al., 2012). The degree of anisotropy is very low (Fig. 4c) and the mean anisotropy is only 1.03 ± 0.03.

4.2. Paleomagnetic results

The intensities of the natural remanent magnetization (NRM) of the majority of samples vary from 2.93×10^{-4} A/m to 8.70×10^{-3} A/m. About one third of specimens yield erratic directions upon demagnetization and are excluded from further analysis. Most of the remaining specimens exhibit two-component magnetizations (Fig. 5a–f) and granitic pebbles/cobbles from the conglomerate beds often show one-component magnetization, with an unblocking temperature at ~580 °C (Fig. 5g). For the two-component magnetizations, the low temperature component (LTC) is typically removed by ~300 °C or ~30 mT, and the high temperature component (HTC) is generally isolated from about 300 °C up to 570–600 °C or from 30 mT to 90–100 mT. The LTC directions are well concentrated in geographic coordinates with a site-mean of $D_g = 355.8°$, $I_g = 62.6°$, $\alpha_{95} = 3.0°$, which is similar to the present Earth's field (PEF) and is interpreted as the present-day overprint. The HTCs decay toward the origin (Fig. 5a–f) and represent the ChRMs of specimens. All ChRMs show normal polarities except sites AK12 and AK13 that contain ChRMs of reversed polarities. The site mean directions are summarized in Table 1.

4.3. Petrographic results

Examination of polished thin sections with a microscope reveals that the studied sandstones are composed of ~35–55% quartz (100–800 μm), ~30–50% feldspar (100–1500 μm), ~13–15% lithic fragments (100–1300 μm), and rare heavy minerals such as epidote, sphene, biotite, muscovite, zircon and opaque minerals (Fig. 6a–d). The greenish siltstones are dominated by fine-grained monocrystalline quartz and plagioclase (Fig. 6d). Boundaries of grains and lithic fragments, particularly opaque phases probably including magnetite in both sandstones and siltstones, are sharply defined and show no signs of mineralogic alterations. These features suggest that the studied rocks are pristine.

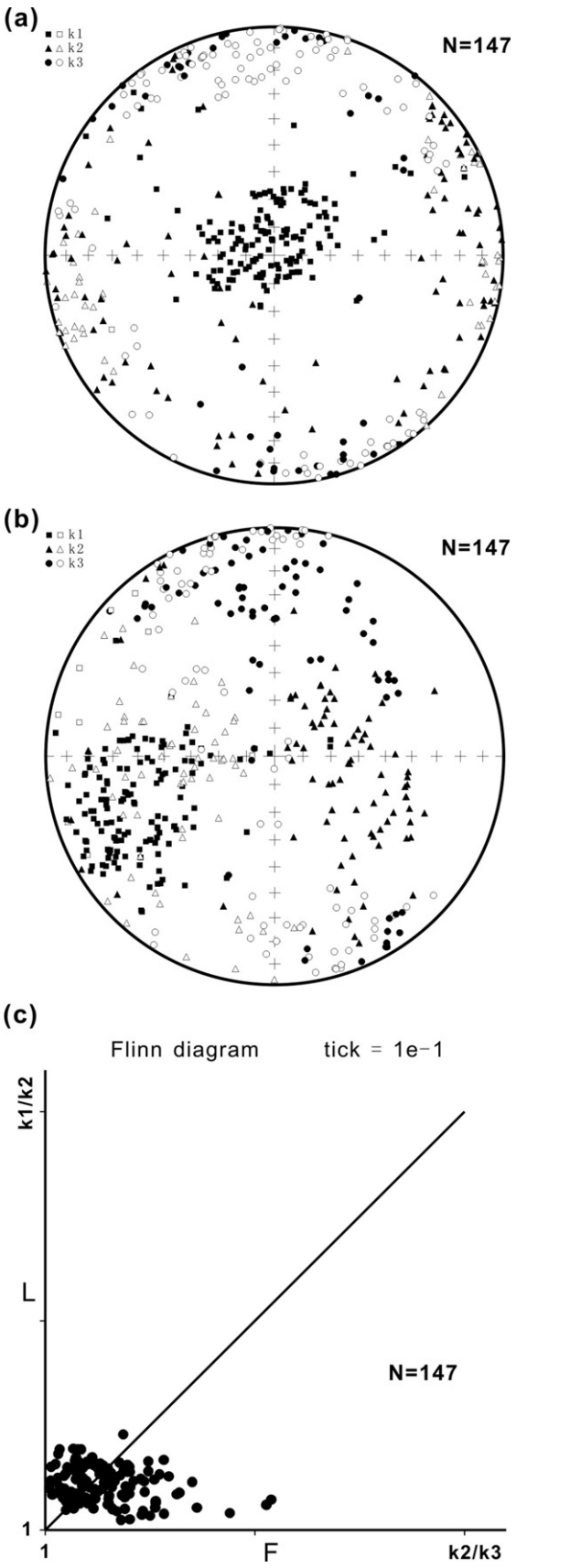


Fig. 4. Anisotropy of magnetic susceptibility (AMS) data of samples are shown in geographic (a) and stratigraphic coordinates (b), and in a Flinn diagram (c) (Flinn, 1962). N, number of samples measured; F, foliation; L, lineation.

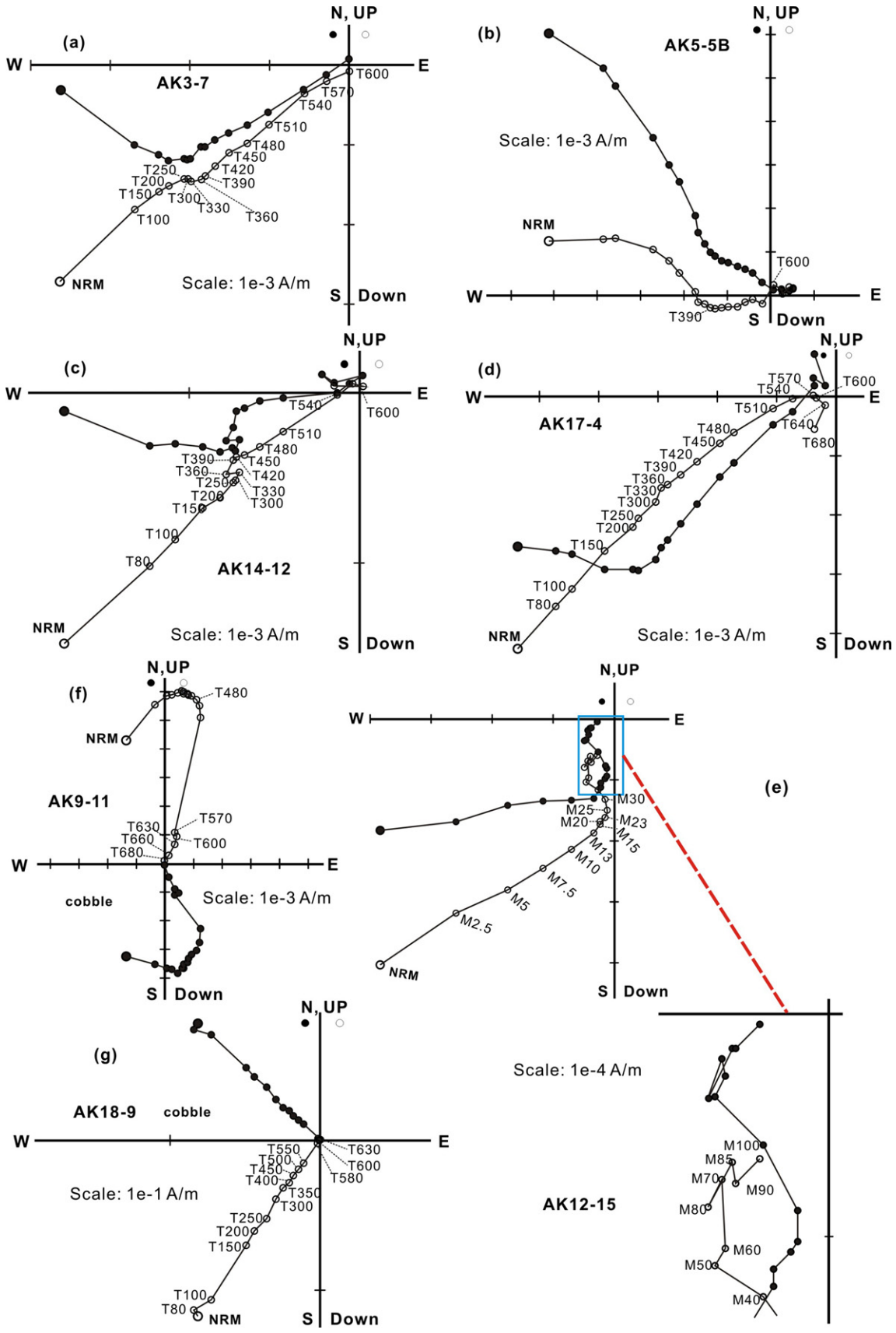


Fig. 5. Progressive demagnetization plots of representative samples in stratigraphic coordinates. Solid (open) circles show projections in the horizontal (vertical) plane. T, M denotes thermal (°C) and alternating filed (mT) demagnetization, respectively.

Table 1
Summary of paleomagnetic results from the Qiaoenbrak Fm in the Aksu area, NW China.

Section	Site	Lithology	S-Lat	S-Long	Strike	Dip	n/n_0	Polarity	D_g	I_g	D_s	I_s	k	α_{95}/MAD	P-pole	P-lat
Strata overlying the diamictites																
	AK20	Red sandstone	40°55′21″	79°27′38″	149	60	8/9	N	46.2	72.2	244.0	46.9	22.1	12.0		
	AK19	Greenish sandstone	40°55′21″	79°27′38″	149	60	1/11	N	323.4	71.6	259.0	26.4	–	/3.6		
	AK15	Greenish sandstone	40°55′24″	79°27′38″	149	60	7/12	N	79.8	77.6	232.6	41.0	35.6	10.3		
	AK14	Greenish sandstone	40°55′22″	79°27′38″	149	60	6/12	N	352.8	84.1	244.8	31.9	20.0	15.4		
	AK13	Sandstone	40°55′21″	79°27′39″	149	60	1/11	R	355.9	−72.4	41.9	−20.5	–	/4.6		
	AK12[a]	Sandstone	40°55′22″	79°27′39″	149	60	3/15	N+R	94.4	32.9	164.0	59.2	38.0	20.3		
Diamictites																
	AK11	Interbedded sandstone	40°55′22″	79°27′40″	149	60	0/10	–	–	–	–	–	–	–		
Upper C.B.	AK10	Interbedded sandstone	40°55′21″	79°27′40″	160	53	7/13	N	309.9	64.2	273.9	21.5	95.3	6.2		
	AK18	Granite, sandstone (pebbles)	40°55′22″	79°27′40″	160	53	6/8	N+R	(conglomerate test)							
	AK17	Thick silt- and mud-stone	40°55′22″	79°27′41″	160	53	7/11	N	199.3	63.8	228.9	18.4	68.8	7.3		
	AK16	Silt- and mud-stone (fold)	40°55′24″	79°27′41″	160	53	9/10	N+R	(soft-sediment fold test)							
Lower C.B.	AK9	Granite, sandstone (pebbles)	40°55′23″	79°27′43″	160	53	8/11	N+R	(conglomerate test)							
Strata underlying the diamictites																
	AK8	Coarse sandstone	40°55′24″	79°27′43″	160	53	4/8	N	0.4	59.9	289.3	42.1	36.4	15.4		
	AK7	Red sandstone	40°55′25″	79°27′43″	160	53	4/10	N	250.6	66.1	250.2	14.1	46.4	13.6		
	AK6	Greenish sandstone	40°55′25″	79°27′44″	160	53	8/8	N	266.1	79.3	253.3	27.7	66.3	6.9		
	AK5	Red sandstone	40°55′25″	79°27′44″	160	53	3/8	N	229.9	79.3	245.8	27.9	49.5	17.7		
	AK4	Red sandstone	40°55′25″	79°27′44″	160	53	11/12	N	326.4	75.7	266.7	33.1	30.6	8.4		
	AK3	Red sandstone	40°55′24″	79°27′44″	160	53	10/11	N	140.0	82.9	241.4	39.1	181.3	3.6		
	AK2	Red sandstone	40°55′24″	79°27′45″	160	53	1/11	N	179.4	41.7	204.0	11.7	–	/8.2		
	AK1	Red sandstone	40°55′25″	79°27′45″	160	53	0/10	–	–	–	–	–	–	–		
$Mean_{14}$ (14 sites above, i.e. Ak2–8, 10, 13–15, 17, 19, 20)							14	N+R	256.6	85.6			13.5	11.3		
											246.1	30.4	14.1	11.0	6.3°S, 17.5°E	
															A_{95} = 9.1	16.3±5.6°
[b]$Mean_{11}$ (11sites that have multiple ChRMs, i.e.14 sites minus Ak2, 13, 19)							11	N	304.2	84.5			19.0	10.7		
											251.6	32.4	19.8	10.5	1.7°S, 14.7°E	
															A_{95} = 8.9	17.6±5.3°

Notes: C.B. = conglomerate bed; S-Lat/S-Long, latitude and longitude of a sampling site; n/n_0, number of specimens in calculation/number of specimens demagnetized; N and R, normal and reversed polarities; D_g, I_g, D_s and I_s, declination and inclination in geographic and stratigraphic coordinates, respectively; k, precision parameter; α_{95}, 95% confidence circle; MAD, maximum angular deviation; P-pole/P-lat, paleomagnetic pole and paleolatitude. Strike/Dip, the strike and dip of the strata following the right hand rule. All angles are in degrees.

[a] Sites that have α_{95} greater than 20° are excluded for calculating the means.

[b] See Section 5.1.3 for details.

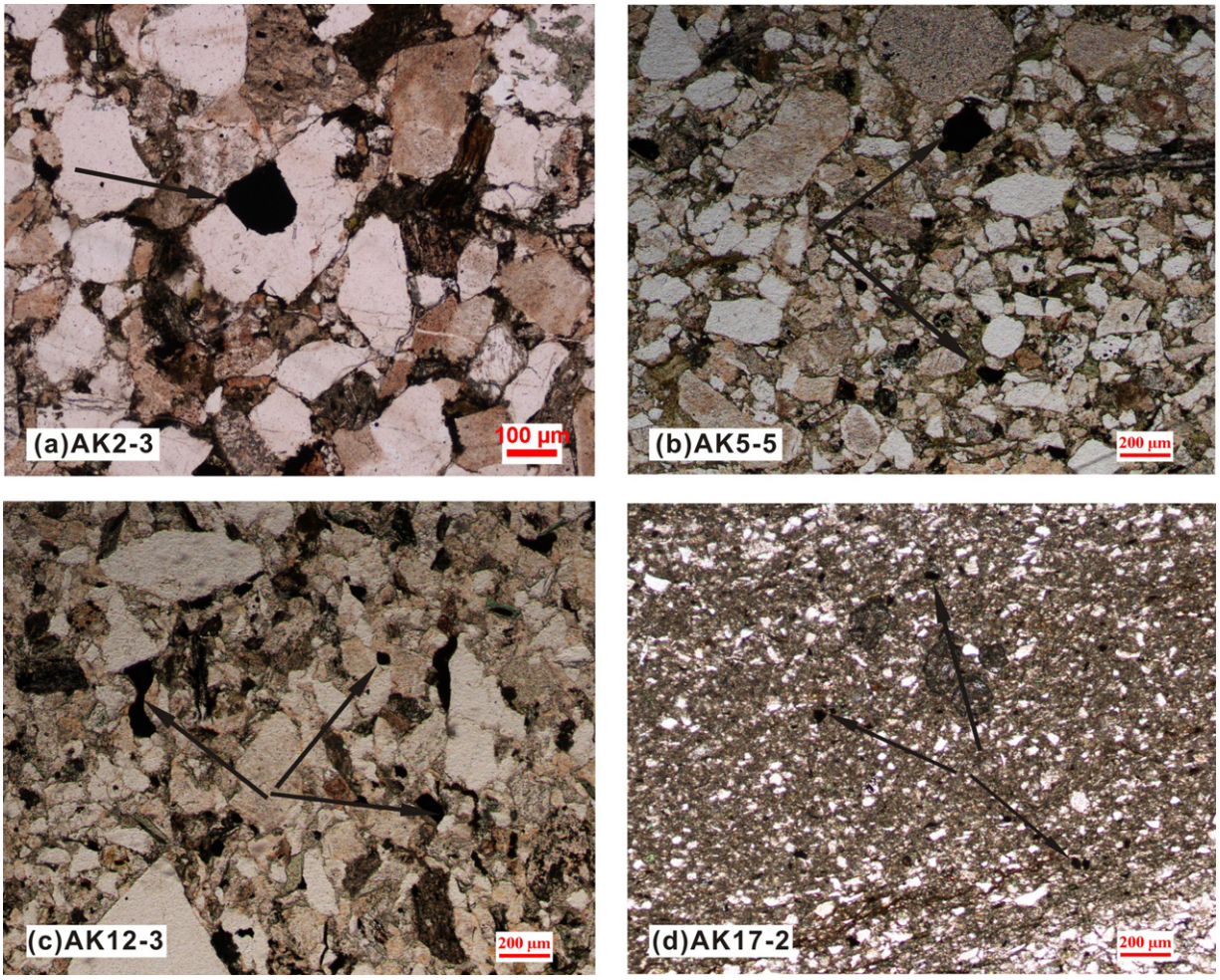


Fig. 6. Photomicrographs of thin sections of the studied rocks show pristine sedimentary features without mineralogic alterations. The opaque minerals including magnetite are marked with arrows.

5. Discussion

5.1. Timing of the remanence acquisition

5.1.1. Field stability tests

The age of the ChRMs is constrained by a fold test and two conglomerate tests. Nine out of 10 samples from both limbs and the hinge of the isoclinal, soft-sediment fold within the greenish siltstones (AK16, Fig. 1c) yield stable ChRMs (Table 2). The ChRMs are moderately grouped in geographic coordinates (Fig. 7a), and become dispersed (Fig. 7b) after the tilt correction with the k_g/k_s ratio of 24.1/1.4. A fold test indicates that the maximum grouping occurs at $4.64 \pm 6.36\%$ untilting (Fig. 7c), suggesting a negative fold test (Enkin, 2003). Consequently, the remanence was acquired after the soft-sediment folding.

The randomness test (Watson, 1956) was applied to the two conglomerate beds that are separated by the greenish siltstone sequence. Eight out of 11 samples from the lower conglomerate bed yield ChRMs (Fig. 8a) and the resultant vector of the ChRMs is $R = 4.12$, which is less than the critical value of $R_0 = 4.48$ for $N = 8$ at 95% confidence level (Watson, 1956), indicating a passage of the conglomerate test. For the upper conglomerate bed, six out of 8 samples yield stable remanent magnetizations (Fig. 8b) and the resultant vector $R = 3.76$, which is less than 3.85, the critical value

Table 2
ChRMs of samples from the isoclinal, soft-sediment fold (site AK16) in the Qiaoenbrak Fm.

Limb	Sample	Strike	Dip	D_g	I_g	D_s	I_s	MAD
Left	AK16-1	143	53	259.7	66.4	243.7	15.4	3.7
	AK16-2	143	53	232.2	69.0	232.7	16.0	3.4
	AK16-3	138	61	288.1	66.5	249.0	15.7	7.9
Right	AK16-4	310	114	282.9	78.5	29.2	−18.4	4.4
	AK16-6	320	130	241.3	63.3	44.8	−13.7	7.5
	AK16-7	320	130	212.5	55.7	59.8	−6.9	7.6
	AK16-8	310	114	203.2	43.8	52.8	20.5	10.9
	AK16-9	310	114	289.8	74.3	24.5	−17.8	7.3
	AK16-10	310	114	240.1	71.2	33.6	−6.2	5.4

Abbreviations: D_g, I_g, D_s and I_s; declination and inclination in geographic and stratigraphic coordinates, respectively; MAD, maximum angular deviation; strike/dip, the strike and dip of the strata following the right hand rule.

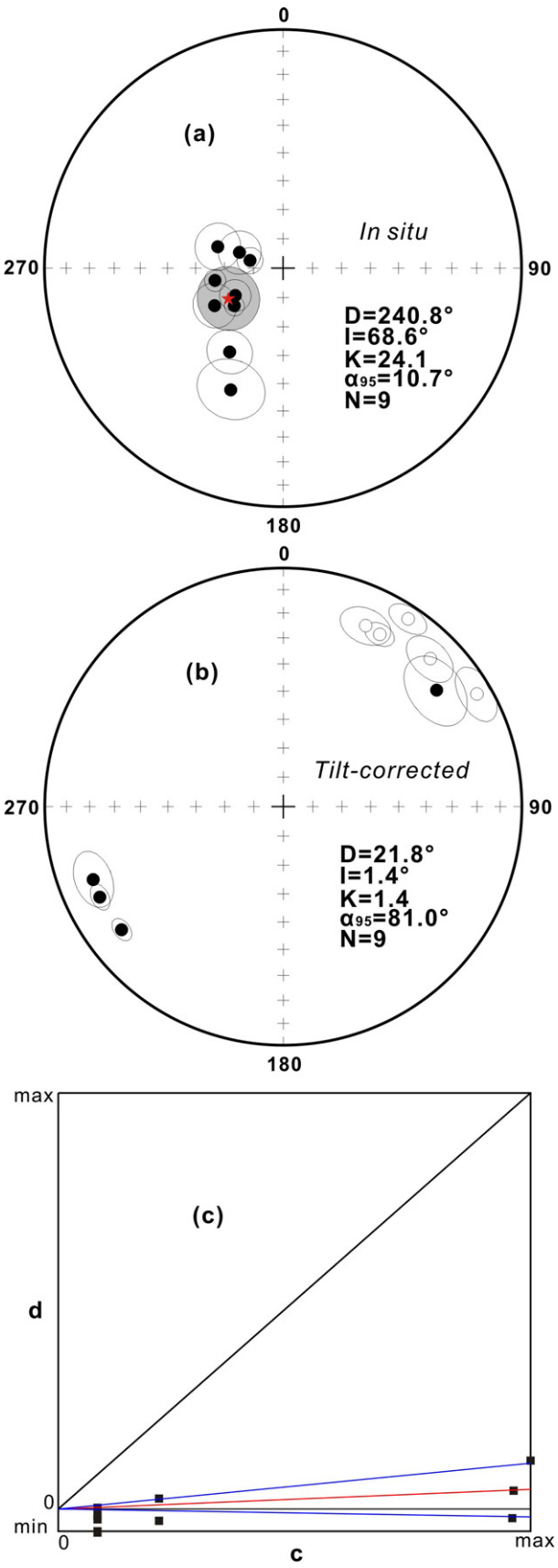


Fig. 7. Direction–Correction Test (Enkin, 2003) of the ChRMs from the soft-sediment fold. The ChRMs are moderately grouped in geographic coordinates (a) and become dispersed after unfolding (b) with the k_g/k_s ratio of 24.1/1.4. Optimal concentration of the ChRMs occurs at 4.64 ± 6.36% untilting (c), indicating a negative fold test (Enkin, 2003).

of R_0 for $N = 6$ at 95% confidence level, constituting a passage of the conglomerate test. The passage of the two conglomerate tests suggests that pebbles/cobbles acquired their magnetization prior to the formation of the diamictites and have retained their original remanence. Although most pebbles/cobbles in the conglomerate beds are granites, which could make the conglomerate test incomplete, polished thin sections of sandstones and siltstones display pristine sedimentary textures and show no signs of mineral alterations (Fig. 6). Taken together, the positive conglomerate tests are regarded complete.

Since soft-sediment folding occurs in the early stage of deposition when water content is modest such that sediments are cohesive enough to retain the shape of a layer and yet soft enough to easily deform. Therefore, the negative soft-sediment fold test and the positive conglomerate tests suggest that the ChRMs likely represent the post-depositional remanent magnetizations (pDRMs), i.e. remanence acquired shortly after deposition (Butler, 1992; Tauxe, 2010). Indeed, pDRM is believed common in many sedimentary environments and is often considered an accurate recorder of the geomagnetic field at or very close to the time of deposition (Butler, 1992).

5.1.2. Age of the diamictite in the Qiaoenbrak Fm

The Qiaoenbrak Fm was considered to be the Mid-Neoproterozoic in age based on the presence of micro-paleobotanic fossils (Gao et al., 1985). The age of the Qiaoenbrak Fm can be further constrained by the radiometric ages obtained from rocks below and above the Qiaoenbrak Fm (Fig. 1b). The Aksu Group, which underlays the Qiaoenbrak Fm, comprises metasedimentary rocks and mafic schists, and was intruded by a series of NW-trending unmetamorphosed mafic dykes. LA-ICP-MS U–Pb dating of detrital zircons from the metasedimentary rocks of the Aksu Group yields a maximum deposition age of 727 ± 12 Ma (Zhu et al., 2011). SHRIMP U–Pb zircon dating of the unmetamorphosed mafic dykes gives an emplacement age of 759 ± 7 Ma (Zhang et al., 2009a). Consequently, a maximum age of ca. 730–760 Ma is inferred for the Qiaoenbrak Fm from these age data. Overlying the Qiaoenbrak Fm is the Yuermeinak Fm, which is covered by the Sugetbrak Fm (Fig. 1b). LA-ICP-MS U–Pb dating of detrital zircons from the uppermost Qiaoenbrak Fm and the lowermost Yuermeinak Fm yields maximum deposition ages of 727 ± 8 Ma and 719 ± 9 Ma, respectively (Ding, H.F., personal communication), which are compatible with the age of ~600 Ma for the Sugertbrak Fm (Zhan et al., 2007; Zhu et al., 2011) that overlays the Yuermeinak Fm (Fig. 1b). These ages allow a minimum age for the Qiaoenbrak Fm to be placed at ~720 Ma. Therefore, the age of the Qiaoenbrak Fm can be bracketed between ~720 and ~760 Ma based on the U–Pb zircon age data from the Aksu area.

The age of the Qiaoenbrak Fm, particularly the diamictites, can be further constrained by the ages of its equivalent Neoproterozoic strata from other parts of the Tarim Block. It is well known that the Neoproterozoic strata in the Aksu area can be correlated with those in the Quruqtagh area at the northeastern margin of the Tarim Block (Fig. 1a). In particular, the diamictites in the Qiaoenbrak Fm in the Aksu area is considered coeval with the diamictites in the Baiyisi Fm in the Quruqtagh area (Fig. 1a; Gao et al., 1981, 1985; XBGMR, 1993). The diamictites in the Baiyisi Fm are sandwiched into two basalt layers. SHRIMP zircon U–Pb dating of the volcanic rocks above the diamictites yields an age of 725 ± 10 Ma (Xu et al., 2009) and the volcanic bed below the diamictites gives two ages of 755 ± 15 Ma (Xu et al., 2005) and 740 ± 7 Ma (Xu et al., 2009). Therefore, the age of diamictites in the Baiyisi Fm is bracketed between ~725 Ma and ~740 Ma. This age interval is compatible with the age of ~720–760 Ma for the Qiaoenbrak Fm, supporting the correlation of diamictites in the Qiaoenbrak Fm and the Bayisi Fm. Therefore, the studied diamictite section in the Aksu area was likely deposited at around 730 Ma.

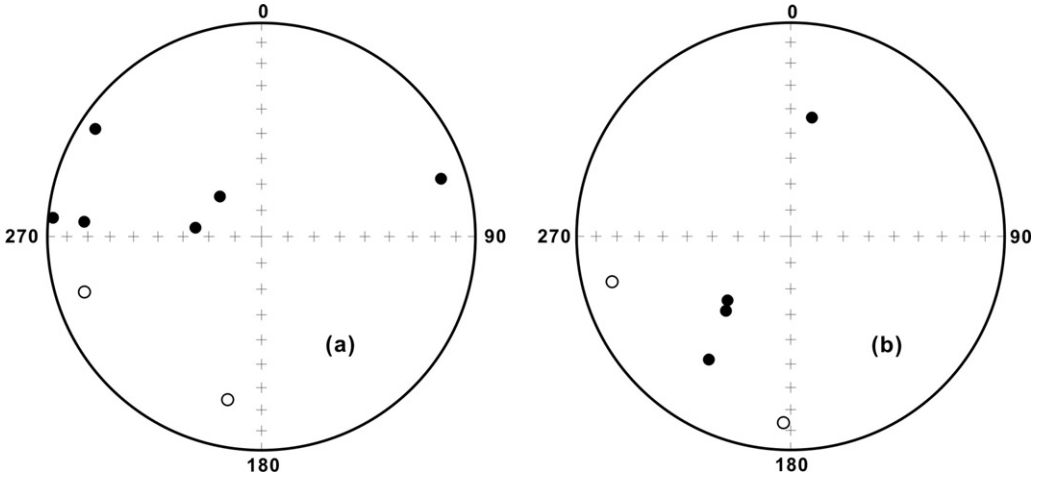


Fig. 8. Equal-area projections of ChRMs of pebbles/cobbles from the lower (a) and upper (b) conglomerate beds of the diamictite unit (Fig. 1c). The ChRMs in each conglomerate bed are randomly distributed, constituting a positive conglomerate test (see Section 5.1.1).

5.1.3. The paleopole of the diamictite in the Qiaoenbrak Fm

Among the 15 sites that yield stable ChRMs and site-means (Table 1), site AK12 shows an α_{95} greater than 20° and is excluded from further analysis. The mean of site-means of the remaining 14 sites is calculated at $D_g = 256.6°$, $I_g = 85.6°$, $k_g = 13.5$, $\alpha_{95} = 11.3°$ in geographic coordinates, and $D_s = 246.1°$, $I_s = 30.4°$, $k_s = 14.1$, $\alpha_{95} = 11.0°$ in stratigraphic coordinates (Fig. 9a and b), which corresponds to a paleopole at $\lambda = 6.3°$S, $\varphi = 17.5°$E, $A_{95} = 9.1°$ with a paleolatitude of $16.3 \pm 5.6°$. This paleopole is significantly different from the previously published younger paleopoles of the Tarim Block (Fig. 10) (Li, 1990; Zhan et al., 2007; Zhao et al., 2008; Gilder et al., 2008; Huang et al., 2005; Sun and Huang, 2009), further attesting the primary origin of the remanence of the studied section.

To test the robustness of this new paleopole, we perform the following analysis. Because sites AK2, AK13, and AK19 has only one stable ChRM at each site and the site mean for sites AK2, AK13 and AK19 is based on only one ChRM at each site (Table 1). These three sites are thus excluded to obtain a new site-mean direction of $D_g = 304.2°$, $I_g = 84.5°$, $k_g = 19.0$, $\alpha_{95} = 10.7°$ in geographic coordinates and $D_s = 251.6°$, $I_s = 32.4°$, $k_s = 19.8$, $\alpha_{95} = 10.5°$ in stratigraphic coordinates (Fig. 9c and d), which corresponds to a new paleopole at $\lambda = 1.7°$S, $\varphi = 14.7°$E, $A_{95} = 8.9°$, with a paleolatitude of $17.6 \pm 5.3°$ (Table 1). Obviously, the two paleopoles and paleolatitudes are essentially same, demonstrating that the paleopole ($\lambda = 6.3°$S, $\varphi = 17.5°$E, $A_{95} = 9.1°$, $N = 14$) with a quality factor $Q = 6$ (Van der Voo, 1990) obtained from the studied section is robust.

5.2. Implications for the late Neoproterozoic paleogeography of the Tarim Block

In paleogeographic reconstructions, similar tectonostratigraphic records serve as the first-order evidence for the close proximity of different blocks. While it is long known that Tarim and Australia have highly comparable Neoproterozoic tectonostratigraphy (Hoffman, 1991; Li et al., 1996, 2008; Zhao and Cawood, 1999; Li and Powell, 2001; Zhao et al., 2004; Lu et al., 2008; Rino et al., 2008; Turner, 2010), the similarity in the Neoproterozoic history between Tarim and South China is relatively recently recognized and is increasingly appreciated (Lu et al., 2003, 2008). The ca. 1.0 Ga Tarimian Orogeny (Zhang et al., 2003) is similar to the Jinningian (or Sibao) Orogeny of South China in character and timing, and both likely corresponded to the Grenville Orogeny during the assembly of Rodinia (Lu et al., 2008). Also, mafic intrusions of similar ages (ca. 820–750 Ma) occur in both Tarim and South China (Table 3), indicating coeval rifting events that were likely associated with the breakup of Rodinia (Li et al., 1999, 2003). In addition, the subsequent Cryogenian (Nanhua) and Edicaran (Sinian) sedimentary sequences, particularly the intervals of glacial deposits in Nanhua System, in Tarim are highly comparable with the Neoproterozoic successions in South China (Lu et al., 2003). The striking similarity in the Neoproterozoic stratigraphy and tectonic history between Tarim and South China strongly suggests that these two blocks were contiguous to each other in Rodinia (Lu et al., 2003, 2008).

Table 3
Summary of radiometric ages of the Neoproterozoic mafic dyke swarms in Tarim, Australia, and South China.

Block	Rock unit	Location	Age (Ma)	Dating method	Reference
Tarim	Aksu dyke swarm (ADS)	Aksu	807 ± 12 Ma	SHRIMP U–Pb	Chen et al. (2004)
	Aksu dyke swarm (ADS)	Aksu	785 ± 30 Ma	SHRIMP U–Pb	Zhan et al. (2007)
	Aksu dyke swarm (ADS)	Aksu	759 ± 7 Ma	SHRIMP U–Pb	Zhang et al. (2009a)
	Quruqtagh dyke swarm	Quruqtagh	773 ± 3 Ma	SHRIMP U–Pb	Zhang et al. (2009a)
	Quruqtagh dyke swarm	Quruqtagh	823–776 Ma	SHRIMP U–Pb	Zhang et al. (2009b)
Australia	Lamprophyre dyke/kimberlite pipes	NW Australia	815–800 Ma	–	Pidgeon et al. (1989)
	Northampton dyke swarm	Western Australia	748 ± 8 Ma	K–Ar	Embleton and Schmidt (1985)
	Mundine Well dyke swarm (MDS)	Western Australia	755 ± 3 Ma	SHRIMP U–Pb	Wingate and Giddings (2000)
	Gairdner Dyke Swarm (GDS)	SE Australia	827 ± 6 Ma	SHRIMP U–Pb	Wingate et al. (1998)
	Amata Dyke swarm	Central Australia	824 ± 4 Ma	–	Sun and Sheraton (1996)
South China	Sanfang dykes	Guangxi	828 ± 7 Ma	SHRIMP U–Pb	Li et al. (1999)
	Kangdian dyke swarm	Sichuan	780–760 Ma	SHRIMP U–Pb	Lin et al. (2007)
	Yanbian dykes	Sichuan	790–760 Ma	SHRIMP U–Pb	Zhu et al. (2008)
	Xiaofeng dykes	Hubei	800 ± 3 Ma	LA-ICP-MS U–Pb	Zhang et al. (2008)

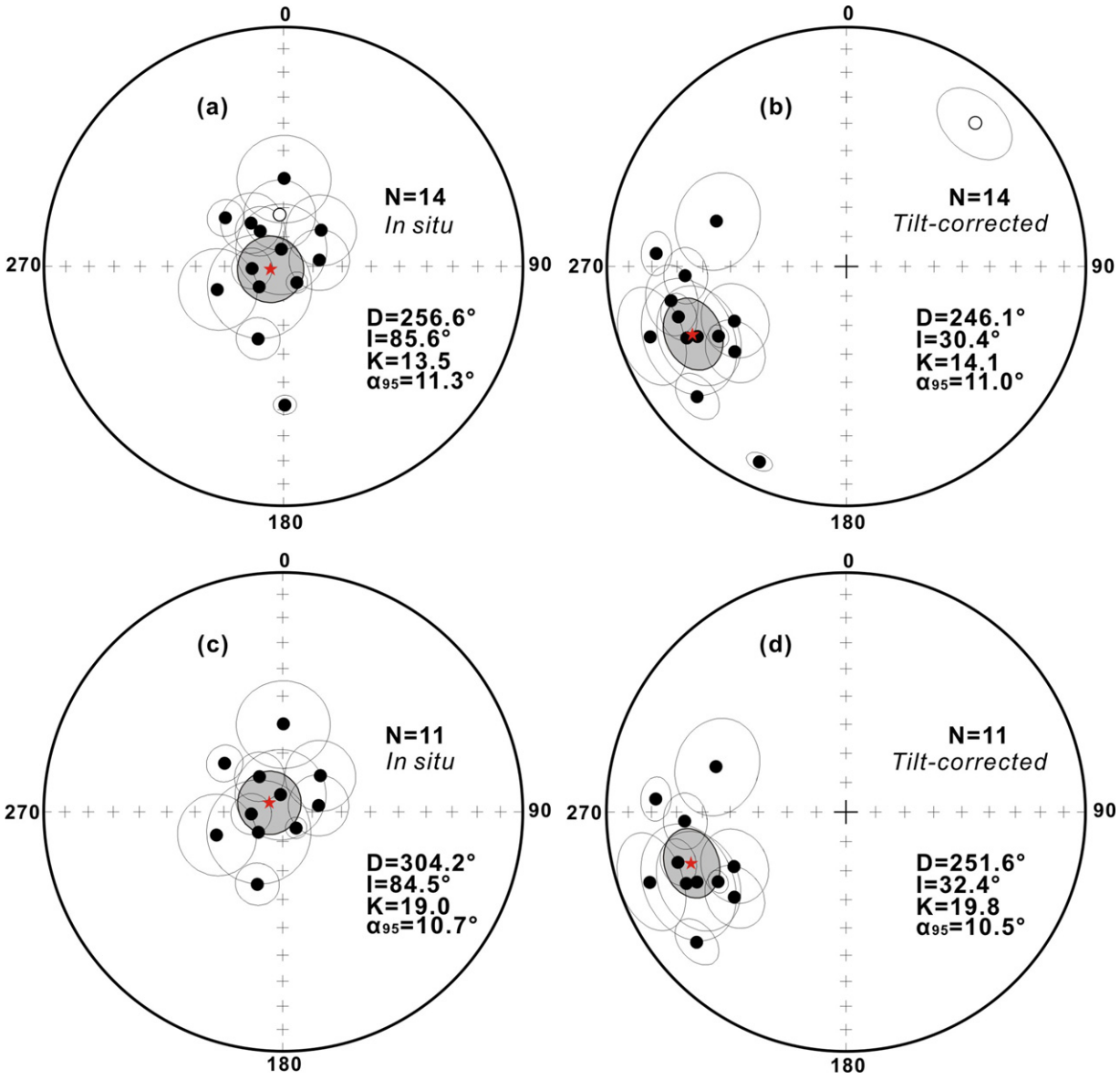


Fig. 9. Equal-area stereographic projections of site-means for $N = 14$ (a and b) and $N = 11$ (c and d), in geographic (a and c) and stratigraphic (b and d) coordinates, respectively. Asterisks represent the mean directions of site means; solid/open circles mark the lower/upper hemispheric projections.

In addition to matching similar tectonostratigraphy, well-dated, reliable paleomagnetic data can provide further quantitative constraints on the positions and orientations of blocks in paleogeographic reconstructions. Rigorous analyses of available paleomagnetic data from various major continents allowed a reconstruction of the ~750 Ma Rodinia (Meert, 2003; Torsvik, 2003), within which Tarim Block is absent probably due to the lack of adequate constraints by high-quality paleomagnetic data. Our reconstruction of the paleogeographic position of Tarim Block at 730 Ma is based mainly on the basic framework of the 750 Ma Rodinia configuration of Torsvik (2003). The approximate positions of major blocks such as Australia and South China at 730 Ma are reconstructed according to their interpolated paleolatitudes using their ~750 Ma paleopoles and the next available good-quality younger paleopoles (Fig. 11). For example, Australia at ~730 Ma is positioned at the interpolated paleolatitude of ~15°N using the ~755 Ma paleopole (Wingate and Giddings, 2000) and the ~600 Ma paleopole (Sohl et al., 1999). Similarly, South China Block is placed at 31°N using the ~748 Ma paleopole (Evans et al., 2000) and the ~630 Ma paleopole (Macouin et al., 2004). Laurentia, India, and other continents are included in Fig. 11 for reference.

Our new paleomagnetic data from the Qiaoenbrak diamictites indicate that Tarim block underwent ~250° clockwise rotations and was located at a low paleolatitude of $16.3 \pm 5.6°$ at around 730 Ma. The paleogeographic position of Tarim at ~730 Ma can then be reconstructed as follows. First, Tarim block is rotated ~250° in a counterclockwise sense to restore its original orientation. Second, geological constraints are utilized to overcome the polarity ambiguities. Since Tarim was likely adjacent to Australia during most of the Neoproterozoic times as evidenced by the marked similarity in tectonostratigraphy between Tarim and Australia (Hoffman, 1991; Li et al., 1996, 2008; Li and Powell, 2001; Lu et al., 2008; Rino et al., 2008; Turner, 2010), and Australia remained in the northerly intermediate to low latitudes throughout the Neoproterozoic time (Sohl et al., 1999; Wingate and Giddings, 2000), Tarim Block is placed at $16.3 \pm 5.6°$N, rather than $16.3 \pm 5.6°$S. Third, as paleomagnetic data cannot constrain the paleolongitude, Tarim Block can be juxtaposed to either the west (model #1) or east (model #2) of Australia at $16.3 \pm 5.6°$N at ca. 730 Ma (Fig. 11). In light of the marked similarity

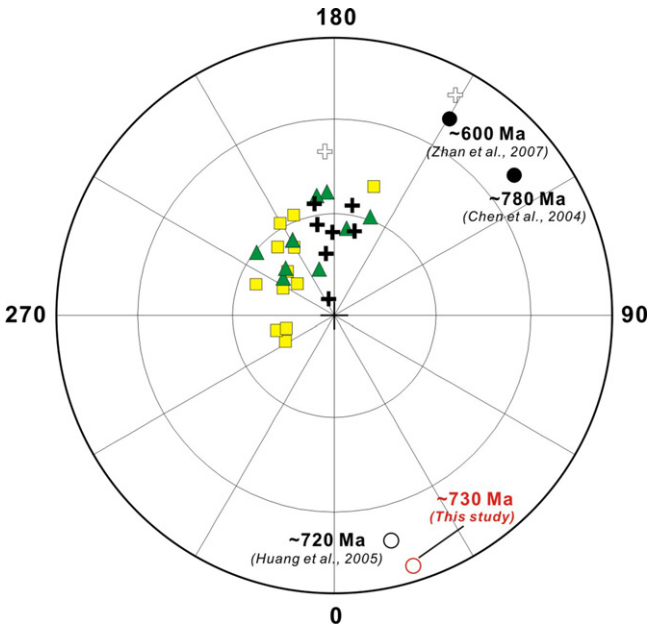


Fig. 10. Comparison of the ~730 Ma paleomagnetic pole of this study with the published Late Neoproterozoic-Cenozoic paleopoles of the Tarim Block (data from Li, 1990; Chen et al., 2004; Huang et al., 2005; Zhan et al., 2007; Gilder et al., 2008; Sun and Huang, 2009). ●,✚, ▲,■, represent the Neoproterozoic, Paleozoic, Mesozoic, and Cenozoic paleopoles, respectively. Solid/open symbols mark the Northern/Southern hemispheric poles.

in the Neoproterozoic history of Tarim and SCB and because of their relatively small sizes, it is feasible to pair up Tarim and SCB in paleogeographic reconstructions to achieve more stringent constraints on the paleogeographic positions of these small blocks than does an individual block alone.

Evidence supporting the "west of Australia" position (WOA; model #1) for the Tarim Block includes the following: (1) the Neoproterzoic strata in the Tarim Block and Australia can be matched (Li et al., 1996, 2008; Turner, 2010), particularly for the three or four glacial intervals between ~750 Ma and ~580 Ma in both cratons (Gao et al., 1985; XBGMR, 1993; Xiao et al., 2004; Trindade and Macouin, 2007; Li et al., 2008; Mahan et al., 2010; Zhu and Wang, 2011; Heron, 2012); (2) the ca. 1000 Ma Tarimian (Grenville) Orogeny that occurred along the southern margin of the Tarim Block (Zhang et al., 2003; Lu et al., 2008) may represent the northward continuation of the ~1000 Ma Proto-Pinjarra Orogeny along the western margin of Australia (Myers, 1993; Myers et al., 1996; Fitzsimons, 2003; Cawood and Korsch, 2008; Ksienzyk et al., 2012); (3) the 820–750 Ma ultramafic–mafic intrusions, bimodal volcanic rocks (Xu et al., 2005, 2009; Zhang et al., 2006, 2007, 2009a,b, 2011; Lu et al., 2008) in the Tarim Block can be correlated with the lamprophyre dykes and kimberlite pipes (Pidgeon et al., 1989; Li et al., 2003), mafic dykes in the Pilbara craton (Embleton and Schmidt, 1985; Wingate and Giddings, 2000; Li et al., 2006), and granitic magmatism in the Leeuwin Block (Collins et al., 2003) in western Australia; (4) our newly determined paleolatitude of the Tarim Block containing the Aksu dyke swarms is similar to that ($16.7 \pm 2.5^{\circ}$) of the 755 Ma Mundine Well dyke swarm in the Pilbara Craton of western Australia (Wingate and Giddings, 2000), implying a potential link of the two cratons; (5) the WOA location is compatible with the subsequent evolution of the Tarim Block in the early Paleozoic, which is based on both paleomagnetic and

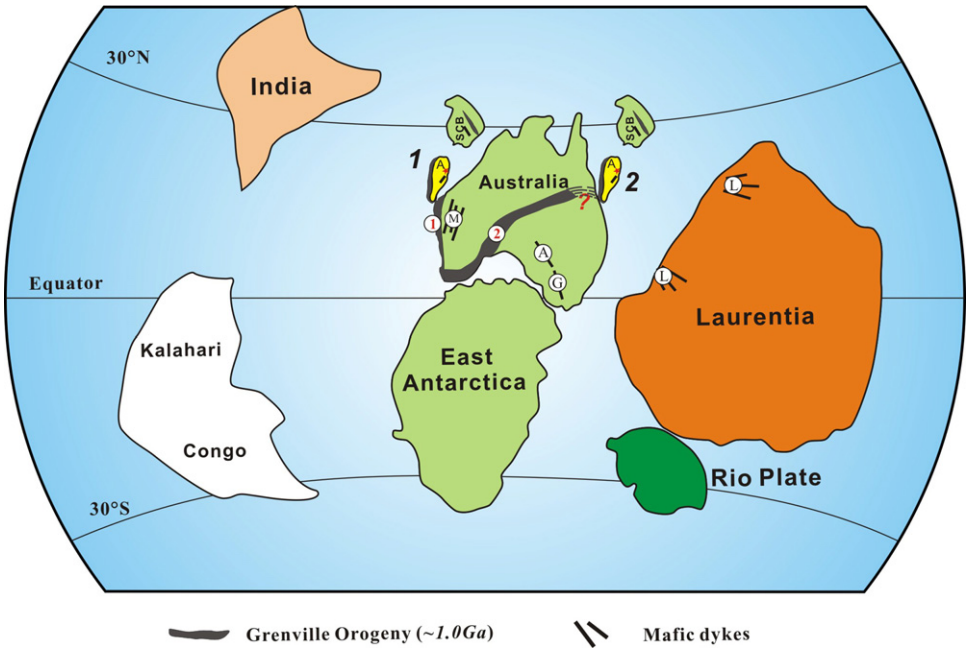


Fig. 11. The Neoproterozoic paleogeographic reconstructions of the Tarim Block at ca. 730 Ma (after Torsvik, 2003; Weil et al., 2004) show two plausible paleogeographic positions (models #1 and #2) for the Tarim Block. A, the Aksu area in the Tarim Block. Grenville Orogen along the present southern margin of the Tarim Block (Zhang et al., 2003; Lu et al., 2008) could be the continuation of the ~1000 Ma Proto-Pinjarra Orogen along the western margin of Australia (Myers, 1993; Myers et al., 1996; Fitzsimons, 2003; Cawood and Korsch, 2008; Ksienzyk et al., 2012) in model #1 or the ~1300–1100 Ma Albany-Fraser-Musgravian Orogen in central Australia (Myers, 1993; Myers et al., 1996; Cawood and Korsch, 2008; Kirkland et al., 2011) in model #2. Mafic dykes in the Tarim Block (Chen et al., 2004; Zhang et al., 2009a,b) may correspond to the Mundine Well dyke swarm (MDS) in the Pilbara Craton of western Australia (Wingate and Giddings, 2000; Li et al., 2006) in model #1 or the mafic dykes in western Laurentia (Park et al., 1995; Li et al., 2008) in model #2. ①, ②, Ⓜ, Ⓖ, Ⓐ, Ⓛ, marks Proto-Pinjarra Orogen and Albany-Fraser-Musgravian Orogen (after Myers et al., 1996), MDS (Wingate and Giddings, 2000), Gairdner Dyke Swarm (GDS; Wingate et al., 1998), Amata dyke swarm and Laurentia mafic dyke swarm (after Li et al., 2003, 2008), respectively. The probable positions of the South China Block (SCB) are shown (after Evans et al., 2000) for reference (Li et al., 1995; Zhang and Piper, 1997; Evans et al., 2000; Li and Powell, 2001; Yang et al., 2004; Li et al., 2008).

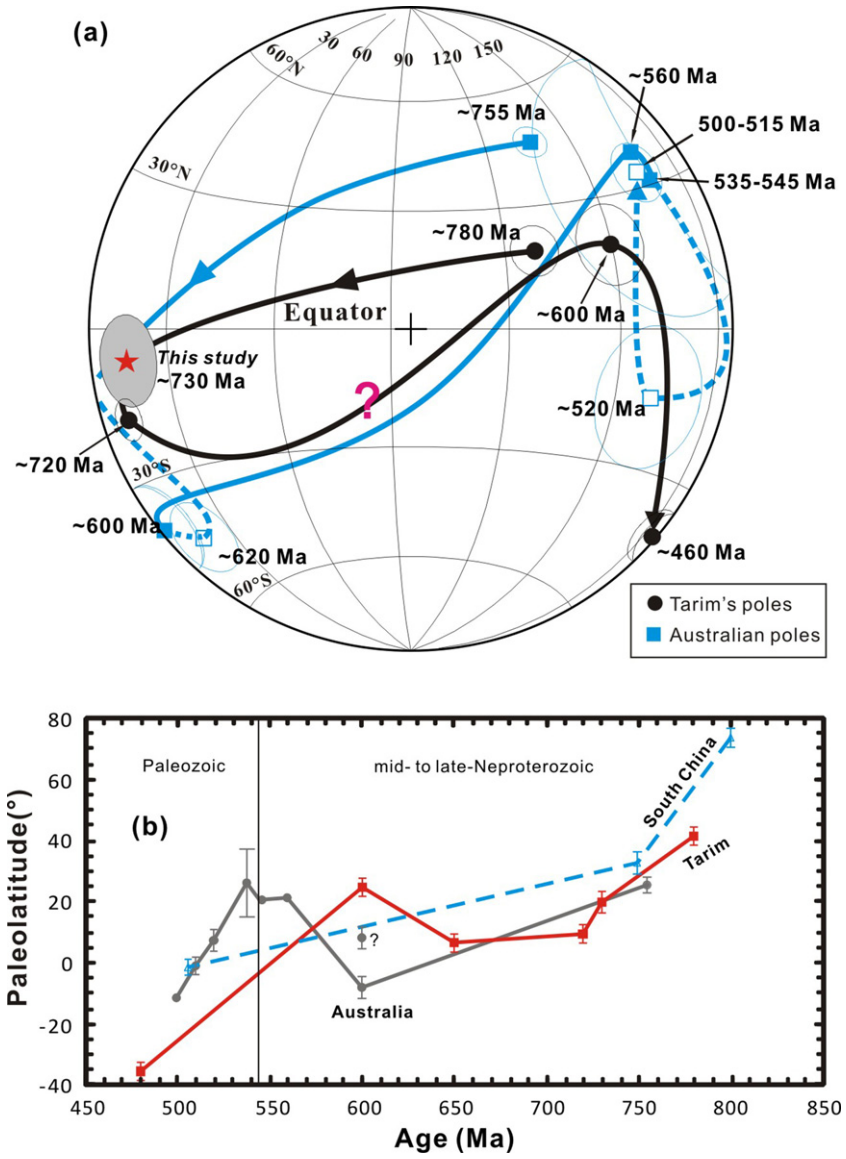


Fig. 12. (a) The preliminary late Neoproterozoic to early Paleozoic apparent polar wander paths (APWPs) of Tarim and Australia (data are provided in Appendix 1). (b) The updated paleolatitude vs. age plot during the time interval for Tarim (reference point at 39°N, 76°E), Australia (−18°N, 123°E), and South China (23°N, 105°E). Vertical bars indicate latitudinal uncertainties at the reference points. The paleolatitudes of Tarim are calculated from a ca. 780 Ma paleopole (Chen et al., 2004), the ca. 730 Ma paleopole of this study, a ca. 720 Ma paleopole (Huang et al., 2005), a ca. 600 Ma paleopole (Zhan et al., 2007), a late-Neoproterozoic paleopole from Wushi (Li et al., 1991), and a newly reported early Paleozoic paleopole (Sun and Huang, 2009). The paleolatitudes of Australia are calculated from a ca. 755 Ma paleopole (Wingate and Giddings, 2000), a key paleopole from Marinoan (~600 Ma) glacial deposits (Elatina Fromation; Sohl et al., 1999), and the latest Neoproterozoic-Cambrian paleopoles summarized by McElhinny et al. (2003). The Paleolatitudes of South China are calculated from a ca. 800 Ma paleopole (Li et al., 2004), a ca. 748 Ma paleopole (Evans et al., 2000), and a middle Cambrian paleopole (Yang et al., 2004).

Bradoriida fossil data (Zhao et al., 2008). The Tarim-South China paired reconstruction for model #1 would place South China close to the northwestern margin of Australia (Fig. 11). This position for South China is plausible because it can satisfy the similar tectonostratigraphic records with Australia and India (e.g. Li et al., 2003; Jiang et al., 2003) and its subsequent tectonic evolution is compatible with that of Australia (e.g. Yang et al., 2004).

It is also possible that the "east of Australia" (EOA; model #2) was linked to the Tarim Block. Besides the similar Neoproterozoic stratigraphy and the glacial intervals between Tarim and Australia, the Grenville Orogen in the Tarim Block may be the eastward continuation of the ~1300–1100 Ma Albany-Fraser-Musgravian Orogen in the central Australia (Myers, 1993; Myers et al., 1996; Cawood and Korsch, 2008; Kirkland et al., 2011). In addition, the ca. 820–720 Ma magmatism in the Tarim Block can be correlated with that in the southeastern Australia (Wingate et al., 1998; Li et al., 2008; Lu et al., 2008; Wang et al., 2010) (Table 3). Furthermore, the rifting event that produced the ~750 Ma (Neoproterozoic) bimodal volcanic rocks in Tarim (Xu et al., 2005, 2009) may correspond to the Franklin event in Laurentia. Together with the similar-aged rifting event in the adjacent South China Block (model #2, Fig. 11) these events could result from a common cause – a mantle plume activity (Heaman et al., 1992; Denyszyn et al., 2009), because the mafic dykes in Tarim, South China, Australia and Laurentia that manifest these rifting events in different continents constitute a giant radiating dyke swarm (Fig. 11), which implies a mantle plume activity. Indeed, similar approach has been employed in recognizing plume activities in not only supercontinent Rodinia (Li et al., 2003), but also other supercontinents such as Columbia (Zhao et al., 2004; Hou et al., 2008a,b). Despite this, model #2 requires a complicated displacement history for the South China Block. The South China Block broke off from the northeastern margin of Australia in

Neoproterozoic. It then had to move from the northeastern margin to the western margin of Australia to account for its physical connection with western Australia in early Paleozoic based on the biogeographic data (e.g. Burrett et al., 1990). However, a driving mechanism for such a required complicated displacement history is lacking.

Overall, the available evidence weighs more toward the WOA position than the EOA position. The WOA position is different from the previous reconstructions that placed the Tarim Block to either the northern Australia (Chen et al., 2004; Zhan et al., 2007) or the southwestern Australia (Huang et al., 2005). The northern Australia position (Chen et al., 2004; Zhan et al., 2007) is mainly based on a proposition that the Aksu dyke swarms in the Tarim Block, which were estimated to be ca. 807 Ma (Chen et al., 2004), represent the northward continuation of the ca. 820–800 Ma lamprophyre dykes and kimberlite pipes (Pidgeon et al., 1989; Li et al., 2003) in the northern Australia. However, recent radiometric data show that the Aksu dyke swarms likely emplaced at 759 ± 7 Ma (SHRIMP U–Pb age; Zhang et al., 2009a) or even younger than ca.730 Ma (Zhu et al., 2011). The disparate ages of Aksu dyke swarms in the Tarim Block and the lamprophyre dykes and kimberlite pipes in the northern Australia may cast doubt on the validity of the early proposition. The WOA position of the Tarim Block appears to be more compatible with the southwestern Australia reconstruction (Huang et al., 2005). Huang et al. (2005) obtained an equatorial paleolatitude for the Tarim Block from the Baiyisi volcanic rocks that were estimated to be ca. 740 Ma. Recent SHRIMP zircon U–Pb geochronologic data from the volcanic rocks overlying and underlying the Baiyisi diamictites show an age of 725 ± 10 Ma and 740 ± 7 Ma, respectively (Xu et al., 2009). Since Huang et al.'s (2005) paleomagnetic data were collected from the volcanic rocks overlying the Baiyisi diamictites, the age of the Baiyisi paleopole would likely be 725 Ma or younger. The WOA position at ~730 Ma would imply that the Tarim Block drifted southward subsequently and reached the equatorial region by ~720 Ma.

Our new paleomagnetic results and previous paleomagnetic data from Tarim Block permit an improved understanding of the kinematics of Tarim Block during the late Neoproterozoic to early Paleozoic interval. In this study, we have obtained a ~730 Ma paleopole at $\lambda = 6.3^{\circ}$S, $\varphi = 17.5^{\circ}$E, $A_{95} = 9.1^{\circ}$, $N = 14$ for Tarim Block. Together with the available late Neoproterozoic to early Paleozoic paleopoles (Chen et al., 2004; Huang et al., 2005; Zhan et al., 2007; Sun and Huang, 2009), a preliminary apparent polar wander path (APWP) for Tarim is reconstructed (Fig. 12a). The overall similar tracks of the APWPs of both blocks suggest the close proximity of Tarim and Australia during the late Neoproterozoic. Tarim Block underwent southward displacement from ~750 Ma to ~720 Ma and remained in the equatorial area from ~720 Ma to 650 Ma. The displacement history of Tarim Block is by and large similar to those of Australia and South China during the interval from ~750 Ma to ~650 Ma (Fig. 12b), suggesting that there were no significant relative latitudinal displacements among these blocks. Tarim Block subsequently moved northward prior to drifting southwardly to reach the intermediate latitudes in the southern hemisphere in the early Paleozoic (Zhao et al., 2008), while both Australia and South China drifted slightly southward and remained in the equatorial region from ~650 Ma to ~600 Ma (Fig. 12b). This indicates that Tarim Block separated from Australia and South China and experienced a different displacement history after ~650 Ma. Additional data are needed to decipher the driving mechanism for the northward displacement of Tarim Block from ~650 Ma to ~600 Ma. Alternatively, Tarim Block may remain close to Australia during the Neoproterozoic–Paleozoic transition because the age of the ~600 Ma paleopole of Tarim Block is not tightly constrained. The age of ~600 Ma of the studied Sugetbrak Fm was estimated by correlating its carbon isotope curve with that of South China (Zhan et al., 2007). The LA-ICP-MS U–Pb dating of detrital zircons from the basal Sugetbrak Fm provided a maximum depositional age of 602 ± 13 Ma for the Sugetbrak Fm (Zhu et al., 2011). Thus, the paleomagnetically studied section of the Sugetbrak Fm (Zhan et al., 2007) is likely younger than 600 Ma. If the age of the studied Sugetbrak section is revised to 560–570 Ma, the latitudinal displacement pattern of Tarim Block would mimic that of Australia during the Neoproterozoic-early Paleozoic transition (Fig. 12b).

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Nevertheless, since the paleomagnetic data of this study were obtained directly from the Qiaoenbrak diamictites, which are of probable glaciogenic origin and equivalent to the Baiyisi diamictites (Gao et al., 1981, 1985; XBGMR, 1993; Zhu and Wang, 2011), this study potentially provides the first direct constraint on the paleolatitude of the glacial deposits in the Tarim Block. Our newly determined low paleolatitude of 16.3 ± 5.6°N joins the paleomagnetic data from other cratons (e.g. Embleton and Williams, 1986; Schmidt and Williams, 1995; Sohl et al., 1999; Schmidt et al., 2009) in supporting the snowball earth hypothesis (Kirschvink, 1992; Hoffman et al., 1998; Hoffman and Schrag, 2002) that argues for low-latitude glaciations during the Neoproterozoic.

6. Conclusions

We have presented a detailed paleomagnetic and rock magnetic study of the Neoproterozoic diamictites of the Qiaoernbrak Fm in the Aksu area, NW China to quantitatively constrain the paleogeographic positions of the Tarim Block during the transition from Rodinia breakup to Gondwana assembly in the middle to late Neoproterozoic. The age of the studied section is constrained at ~730 Ma based on the stratigraphy and recent zircon U–Pb chronologic data. Progressive alternating field (AF) and thermal demagnetization isolated ChRMs that mainly reside in magnetite and are interpreted as pDRMs on the basis of the negative fold test of a soft-sediment fold, the presence of both normal and reversed polarities, positive conglomerate tests, and the pristine sedimentary features of the studied rocks. Fourteen out of 17 sites yielded stable ChRMs that correspond to a paleopole of $\lambda = 6.3^{\circ}$S, $\varphi = 17.5^{\circ}$E, $A_{95} = 9.1^{\circ}$, $N = 14$ with a paleolatitude of 16.3 ± 5.6°. The new paleomagnetic results in conjunction with geologic constraints place the Tarim Block alongside to either west or east of Australia (WOA or EOA) at 16.3 ± 5.6°N and the WOA position is considered more likely based on the collectively favorable evidence. Since the paleomagnetic data are obtained directly from the ~730 Ma diamictites that are probably glaciogenic in origin, the newly determined low paleolatitude provides supportive evidence for the Neoproterozoic snowball earth hypothesis.

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